

SPECIFICATION

Revision: 0.0

This module uses ROHS material

This specification may change without prior notice in order to improve performance or quality. Please contact WKS R&D department for updated specification and product status before design for this product or release of this order.

WRITTEN BY	CHECKED BY	APPROVED BY

REVISION RECORD

REV NO.	REV DATE	CONTENTS	REMARKS
0.0	2022-8-20	First release	Preliminary

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1、GENERAL INFORMATION

Item of general information	Contents	Unit
LCD Display Size (Diagonal)	7.0	inch
LCD Display Type	TFT/TRANSMISSIVE	-
LCD Display Mode	Normally White	-
Recommended Viewing Direction	12	o'clock
Gray inversion Direction	6	o'clock
Module size (W×H×T)	164.90×100.00×5.90	mm
Active area (W×H)	154.08×85.92	mm
Number of pixels (Resolution)	800(RGB)×480	pixel
Pixel pitch (W×H)	0.1926×0.179	mm
Color Pixel Arrangement	RGB Stripe	-
LCD Driver IC	-	-
Interface Type	24bit Parallel RGB interface	-
Color Numbers	16.7M	-
Backlight Type	White LED	-

2、EXTERNAL DIMENSIONS

3、ABSOLUTE MAXIMUM RATINGS

Parameter of absolute maximum ratings	Symbol	Min	Max	Unit
LCD supply voltage	VDD	-0.5	5.0	V
Operating temperature	Top	-20	70	°C
Storage temperature	Tst	-30	80	°C
Humidity	RH	-	90%(Max 60°C)	RH

Note: Absolute maximum ratings means the product can withstand short-term, not more than 120 hours. If it is a long time to withstand these conditions, the life time would be shorter.

4、ELECTRICAL CHARACTERISTICS(DC CHARACTERISTICS)

Parameter of DC characteristics	Symbol	Min.	Typ.	Max.	Unit	Note
LCD Digital operating voltage	VDD	3.0	3.3	3.6	V	
Analog Supply Voltage	AVDD	-	9.6	-	V	
Gate On Voltage	VGH	-	18	-	V	
Gate Off Voltage	VGL	-	-6.8	-	V	
Common Voltage	VCOM	-	3.6	-	V	Note1
Input voltage 'H' level	VIH	0.7*VDD	-	VDD	V	
Input voltage 'L' level	VIL	VSS	-	0.3*VDD	V	
Output voltage 'H' level	VOH	VDD-0.4	-	VDD	V	
Output voltage 'L' level	VOL	VSS	-	VSS+0.4	V	
Digital current	I_{VDD}	---	10	---	mA	
Analog current	I_{AVDD}	---	30	---	mA	
Gate On current	I_{VGH}	---	0.5	---	mA	
Gate Off current	I_{VGL}	---	0.5	---	mA	

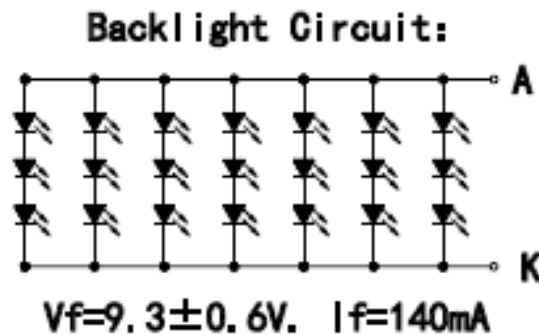
Note 1: Please adjust VCOM voltage to make the flicker level be minimum.

5、BACKLIGHT CHARACTERISTICS

Item of backlight characteristics	Symbol	Min.	Typ.	Max.	Unit	Condition
Forward Voltage	Vf	8.7	9.3	9.9	V	Note1
Forward Current	If	-	140	-	mA	-
Number of LED	-	-	3*7=21	-	Piece	-
LED Connection mode	P/S	-	Serial/Parallel	-	-	-
Lifetime of LED	-	-	20000	-	hour	Note2

Note:

- Note1: The LED Supply Voltage is defined by the number of LED at $T_a=25^{\circ}\text{C}$ and $I_f=140\text{mA}$.
- Note2: The LED lifetime define as the estimated time to 50% degradation of initial luminous. The LED lifetime could be decreased if operating I_f is larger than 140mA.
- Backlight circuit:



6、TOUCH CHARACTERISTICS

Item of CTP characteristics	Specification	Unit	Remark
Panel Type	Glass Cover + Glass Sensor	-	-
Resolution	800 × 480	pixel	-
Surface Hardness	≥6H	-	-
Transparency	>82%	-	-
Driver IC	GT911	-	-
Interface Type	I2C	-	-
Support Points	5(Max)	-	-
Supply voltage	3.3	V	-

7、ELECTRO-OPTICAL CHARACTERISTICS

Item of electro-optical characteristics		Symbol	Condition	Min.	Typ.	Max.	Unit	Remark	Note
Response time		Tr+Tf	$\theta=0$ $\phi=0$ Ta=25°C	-	25	50	ms	FIG 1.	4
Contrast Ratio		CR		-	320	-	-	FIG 2.	1
Luminance uniformity		δ WHITE		-	80	-	%	FIG 2.	3
Surface Luminance		Lv		-	300	-	cd/m2	FIG 2.	2
CIE (x, y) chromaticity	White	White x	$\theta=0$ $\phi=0$ Ta=25°C	0.26	0.31	0.36	-	FIG 2.	5
		White y		0.28	0.33	0.38			
	Red	Red x		0.537	0.587	0.637			
		Red y		0.281	0.331	0.381			
	Green	Green x		0.294	0.344	0.394			
		Green y		0.521	0.571	0.621			
	Blue	Blue x		0.096	0.146	0.196			
		Blue y		0.042	0.092	0.142			
Viewing angle range	$\phi=90(12\text{ o'clock})$		CR ≥ 10	-	50	-	deg	FIG 3.	6
	$\phi=270(6\text{ o'clock})$			-	70	-	deg		
	$\phi=0(3\text{ o'clock})$			-	70	-	deg		
	$\phi=180(9\text{ o'clock})$			-	70	-	deg		
NTSC ratio		-	-	-	50	-	%	-	-

Note 1. Contrast Ratio(CR) is defined mathematically by the following formula. For more information see FIG 2.:

$$\text{Contrast Ratio(CR)} = \frac{\text{Average Surface Luminance with all white pixels(P1,P2,P3,P4,P5,P6,P7,P8,P9)}}{\text{Average Surface Luminance with all black pixels(P1,P2,P3,P4,P5,P6,P7,P8,P9)}}$$

Note 2. Surface luminance is the LCD surface from the surface with all pixels displaying white. For more information see FIG 2.

Lv=Average Surface Luminance with all white pixels (P1,P2,P3,P4,P5,P6,P7,P8,P9)

Note 3. The uniformity in surface luminance (δWHITE) is determined by measuring

luminance at each test position 1 through 9, and then dividing the maximum luminance of 9 points luminance by minimum luminance of 9 points luminance. For more information see FIG 2.

$$\delta_{\text{WHITE}} = \frac{\text{Minimum Surface Luminance with all white pixels (P1, P2, P3, P4, P5, P6, P7, P8, P9)}}{\text{Maximum Surface Luminance with all white pixels (P1, P2, P3, P4, P5, P6, P7, P8, P9)}}$$

Note 4. Response time is the time required for the display to transition from White to black(Rise Time, Tr) and from black to white(Decay Time, Tf). For additional information see FIG 1.

Note 5. CIE (x, y) chromaticity ,The x,y value is determined by screen active area position 5. For more information see FIG 2.

Note 6. Viewing angle is the angle at which the contrast ratio is greater than a specific value. For TFT module, the specific value of contrast ratio is 10.The angles are determined for the horizontal or x axis and the vertical or y axis with respect to the z axis which is normal to the LCD surface. For more information see FIG 3.

Note 7. For Viewing angle and response time testing, the testing data is base on Autronic-Melchers's ConoScope. Series Instruments. For contrast ratio, Surface Luminance, Luminance uniformity and CIE, the testing data is base on BM-7 photo detector.

Note 8. For TN type TFT transmissive module, Gray scale reverse occurs in the direction of panel viewing angle.

FIG.1. The definition of Response Time

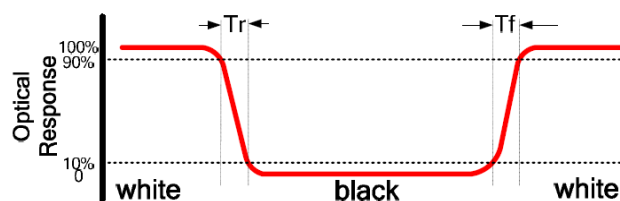


FIG.2. Measuring method for Contrast ratio, surface luminance, Luminance uniformity, CIE (x , y) chromaticity

A : $H/6$;

B : $V/6$;

H,V : Active Area(AA) size

Measurement instrument: BM-7; Light spot size=5mm, 350mm distance from the LCD surface to detector lens.

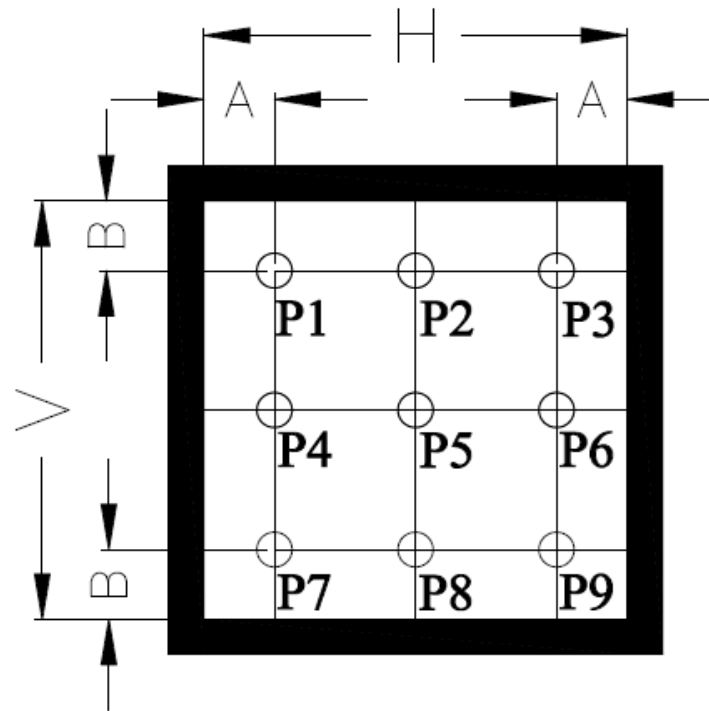
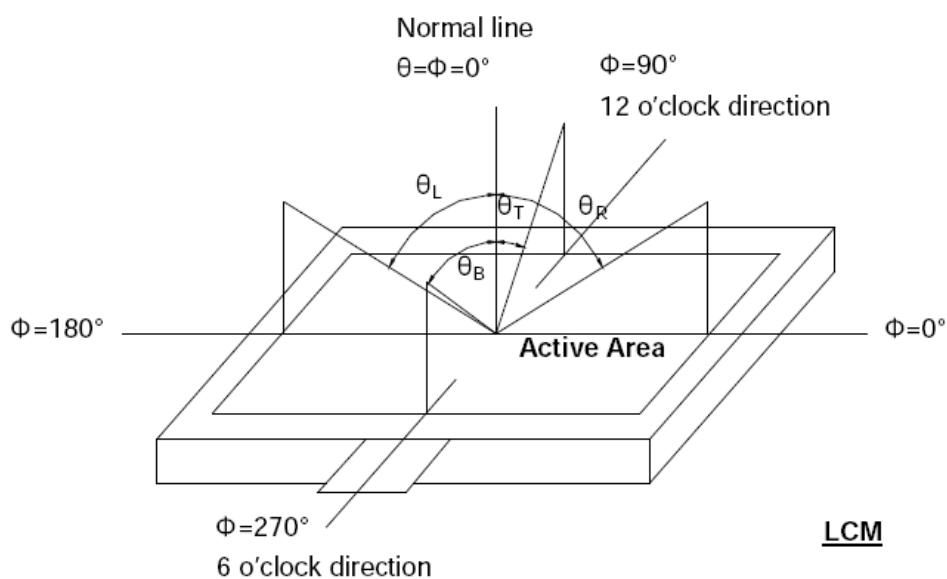


FIG.3. The definition of viewing angle



8、INTERFACE DESCRIPTION

A、LCD Interface Description

NO.	Symbol	I/O	DESCRIPTION
1~2	LED-A	Power supply	Backlight Anode
3~4	LED-K	Power supply	Backlight Cathode
5	GND	Power supply	Power ground
6	VCOM	I	For external VCOM DC input
7	VDD	Power supply	Digital Power supply
8	MODE	I	DE/SYNC mode select. H:DE mode; L:HSD/VSD mode.
9	DEN	I	Data input Enable. Active high to enable the data input Bus.
10	VSYNC	I	Vertical Sync input. Negative polarity.
11	HSYNC	I	Horizontal Sync input. Negative polarity.
12~19	B7~B0	I	8bit digital Blue data input(B7:MSB; B0:LSB)
20~27	G7~G0	I	8bit digital Green data input(G7:MSB; G0:LSB)
28~35	R7~R0	I	8bit digital Red data input(R7:MSB; R0:LSB)
36	GND	Power supply	Power ground
37	PCLK	I	Clock signal. Data latched at rising/falling edge of this signal.
38	GND	Power supply	Power ground
39	SHLR	I	Source Right or Left sequence control.
40	UPDN	I	Gate Up or Down scan control.
41	VGH	Power supply	Positive Power for TFT
42	VGL	Power supply	Negative Power for TFT
43	AVDD	Power supply	Power supply for analog circuits
44	RESET	I	LCD reset signal, Low is active
45	NC	-	No Connection
46	VCOM		For external VCOM DC input
47	DITHB	I	Dithering function enable control
48	GND	Power supply	Power ground
49~50	NC	-	No Connection

B、CTP Interface Description

NO.	Symbol	I/O	DESCRIPTION
1	INT	O	CTP External interrupt to the host
2	SDA	I/O	CTP I2C data input and output
3	SCL	I	CTP I2C clock input
4	RESET	I	CTP external reset signal, Low is active
5	GND	Power supply	Power ground

6	VDD	Power supply	CTP Power input
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9、INPUT TIMING

Horizontal input Timing table

Parameter	Symbol	Value			Unit
		Min.	Typ.	Max.	
DCLK frequency	DCLK	-	33.3	50	MHz
Horizontal display area	thd	800			DCLK
1 Horizontal Line	th	862	1056	1200	DCLK
HSD pulse width	thpw	1	-	40	DCLK
HSD Blanking	thb	46	46	46	DCLK
HSD Front Porch	thfp	16	210	354	DCLK

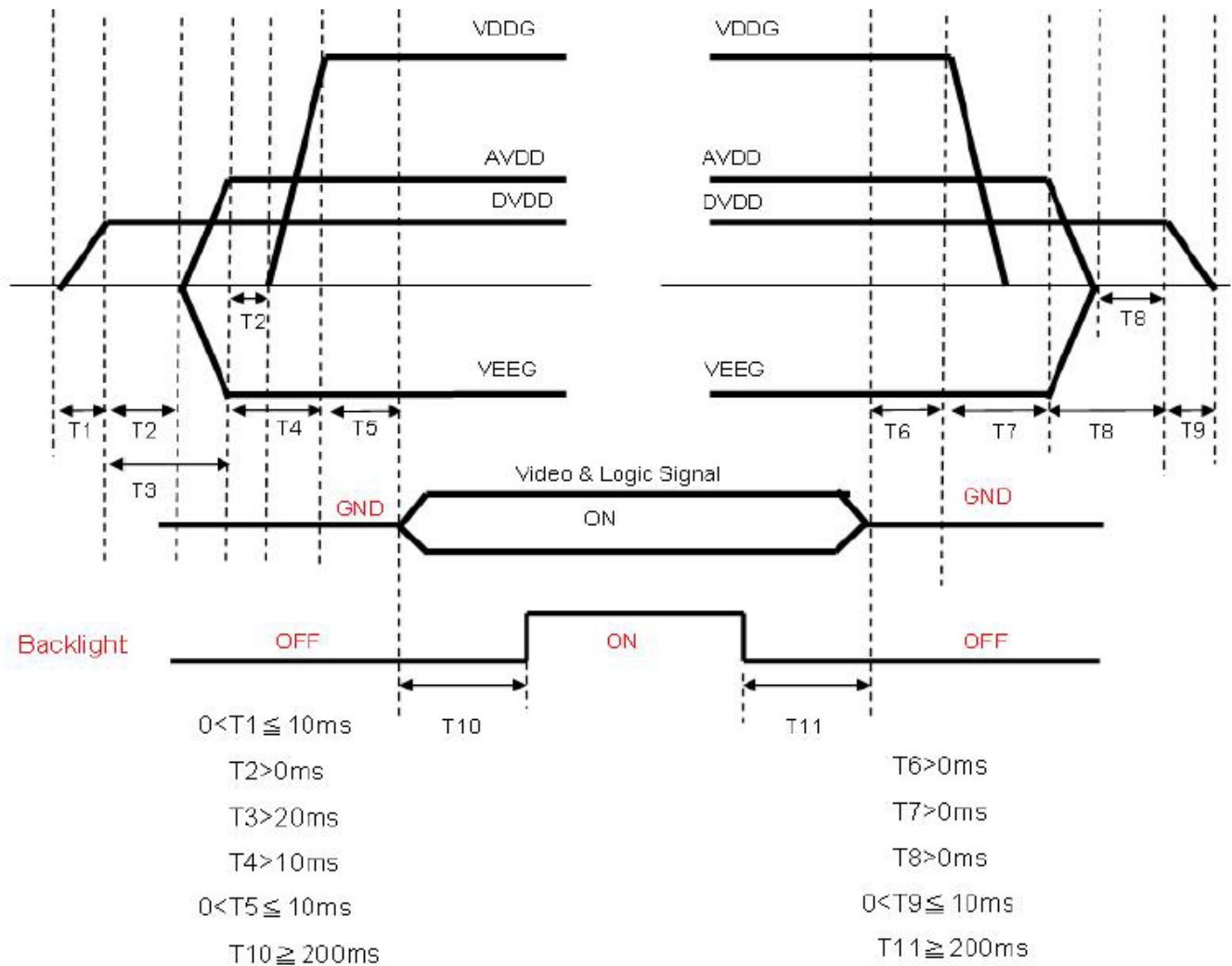
Vertical input Timing table

Parameter	Symbol	Value			Unit
		Min.	Typ.	Max.	
Vertical display area	tvd	480			H
VSD period time	tv	510	525	650	H
VSD pulse width	tvpw	1	-	20	H
VSD Blanking	tvb	23	23	23	H
VSD Front Porch	tvfp	7	22	147	H

10、POWER ON/OFF SEQUENCE

Power On: VDD→AVDD/VGL→VGH→Video & Logic Signal

Power Off: Video & Logic Signal→VGH→AVDD/VGL→VDD



11、RELIABILITY TEST CONDITIONS

No.	Test Item	Test Condition
1	High Temperature Storage	80°C/120 hours
2	Low Temperature Storage	-30°C/120 hours
3	High Temperature Operating	70°C/120 hours
4	Low Temperature Operating	-20°C/120 hours
5	Temperature Cycle Storage	-20°C(30min.)~25(5min.)~70°C(30min.)×10cycles

A、Inspection after test:

Inspection after 2~4 hours storage at room temperature, the sample shall be free from defects:

- Air bubble in the LCD;
- Sealleak;
- Non-display;
- Missing segments;
- Glass crack;
- Current is twice higher than initial value.

B、Remark:

- The test samples should be applied to only one test item.
- Sample size for each test item is 5~10pcs.
- Failure Judgment Criterion: Basic Specification, Electrical Characteristic, Mechanical Characteristic, Optical Characteristic.

12、 INSPECTION CRITERION

This specification is made to be used as the standard of acceptance/rejection criteria for TFT-LCD/IPS TFT-LCD module product, and this specification is applicable only in the case that the size of module equal to or exceed than 3.5 inch.

12.1 Sample plan

Sampling plan according to GB/T2828.1-2003/ISO 2859-1: 1999 and ANSI/ASQC

Z1.4-1993,normal level 2 and based on:

Major defect: AQL 0.65

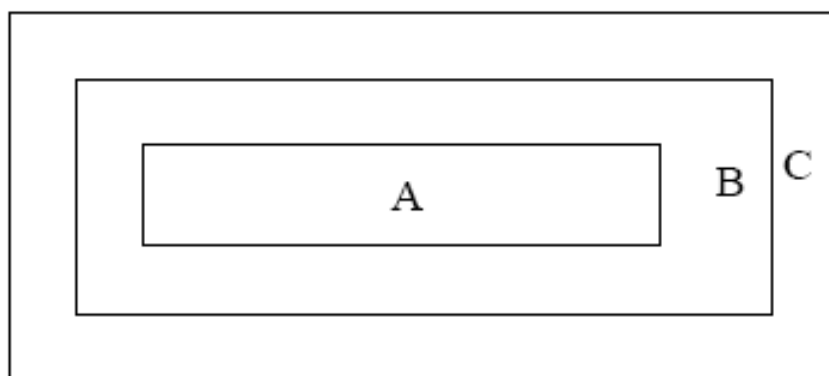
Minor defect: AQL 1.5

12.2 Inspection condition

Viewing distance for cosmetic inspection is about 30cm with bare eyes, and under an environment of 20~40W light intensity, all directions for inspecting the sample should be within 45°against perpendicular line. (Normal temperature 20~25℃and normal humidity 60±15%RH)

12.3 Definition of Inspection Item.

A、 Definition of inspection zone in LCD.



Zone A: character/Digit area

Zone B: viewing area except Zone A (Zone A + Zone B=minimum Viewing area)

Zone C: Outside viewing area (invisible area after assembly in customer's product)

Fig.1 Inspection zones in an LCD

Note: As a general rule, visual defects in Zone C are permissible, when it is no trouble for quality and assembly of customer's product.

B、 Definition of some visual defect

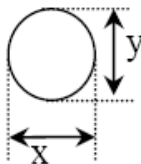
Bright dot	Because of losing all or part function, bad pixel dots appear bright and the size is more than 50% of one dot in which LCD panel is displaying under black pattern.
Dark dot	Dots appear dark and unchanged in size in which LCD panel is displaying under pure red, green, blue picture, or pure whiter picture.

12.4 Major Defect

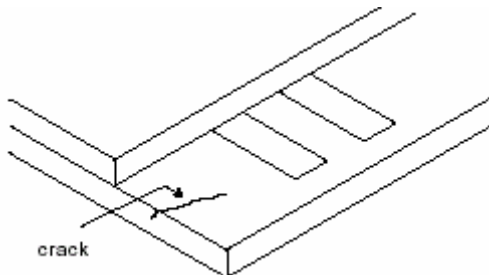
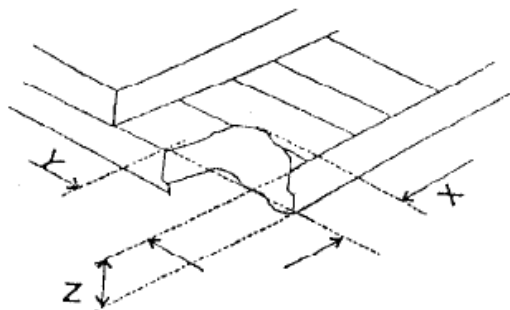
Item No.	Items to be inspected	Inspection standard	Classification of defects
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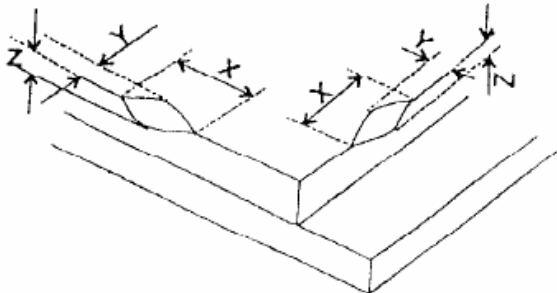
1	Functional defects	1) No display 2) Display abnormally 3) Missing vertical, horizontal segment 4) Short circuit 5) Excess power consumption 6) Backlight no lighting, flickering and abnormal lighting	major
2	Missing	Missing component	
3	Outline dimension	Overall outline dimension beyond the drawing is not allowed	

12.5 Minor Defect

Item No.	Items to be inspected	Inspection standard				Classification of defects																													
1	Bright dot /dark dot defect	<table> <tr> <th rowspan="3">Zone</th> <th colspan="3">Acceptable Qty</th> <th rowspan="3">C</th> </tr> <tr> <th colspan="3">A+B</th> </tr> <tr> <th>3.5~7"</th> <th>7~10.1"</th> <th>>10.1"</th> </tr> <tr> <td>Bright pixel dot</td> <td>1</td> <td>2</td> <td>3</td> <td rowspan="5">Acceptable</td> </tr> <tr> <td>Dark pixel dot</td> <td>4</td> <td>4</td> <td>4</td> </tr> <tr> <td>2bright dots adjacent</td> <td>0</td> <td>0</td> <td>0</td> </tr> <tr> <td>2dark dots adjacent</td> <td>0</td> <td>0</td> <td>0</td> </tr> <tr> <td>Total bright and dark dots</td> <td>5</td> <td>6</td> <td>7</td> </tr> </table> Note: Minimum distance between defective dots is more than 5mm; Pixel dots’ function is normal, but bright dots caused by foreign material and other reasons are judged by the dot defect of 5.2.	Zone	Acceptable Qty			C	A+B			3.5~7"	7~10.1"	>10.1"	Bright pixel dot	1	2	3	Acceptable	Dark pixel dot	4	4	4	2bright dots adjacent	0	0	0	2dark dots adjacent	0	0	0	Total bright and dark dots	5	6	7	Minor
Zone	Acceptable Qty			C																															
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2dark dots adjacent	0	0	0																																
Total bright and dark dots	5	6	7																																
2	Dot defect  $\Phi = (x+y) / 2$	<table> <tr> <th rowspan="3">Size(mm)</th> <th colspan="3">Acceptable Qty</th> <th rowspan="3">C</th> </tr> <tr> <th colspan="3">A+B</th> </tr> <tr> <th>3.5"~7"</th> <th>7~10.1"</th> <th>>10.1"</th> </tr> <tr> <td>$\Phi \leq 0.2$</td> <td>Acceptable</td> <td>Acceptable</td> <td>Acceptable</td> <td rowspan="3">Acceptable</td> </tr> <tr> <td>$0.2 < \Phi \leq 0.5$</td> <td>4</td> <td>5</td> <td>6</td> </tr> <tr> <td>$\Phi > 0.5$</td> <td>0</td> <td>0</td> <td>0</td> </tr> </table> Note: 1. Minimum distance between defective dots is more than 5 mm; 2. The quantity of defect is zero in operating condition.	Size(mm)	Acceptable Qty			C	A+B			3.5"~7"	7~10.1"	>10.1"	$\Phi \leq 0.2$	Acceptable	Acceptable	Acceptable	Acceptable	$0.2 < \Phi \leq 0.5$	4	5	6	$\Phi > 0.5$	0	0	0	Minor								
Size(mm)	Acceptable Qty			C																															
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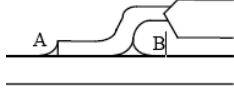
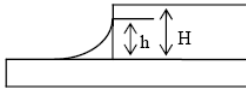
3	Linear defect	<table><tr><th colspan="2" rowspan="2">Zone Size (mm)</th><th colspan="4">Acceptable Qty</th></tr><tr><th colspan="3">A+B</th><th rowspan="4">C</th></tr><tr><th>Length</th><th>Width</th><th>3.5''~7''</th><th>7~10.1''</th><th>>10.1''</th></tr><tr><td>Ignore</td><td>W≤0.05</td><td>Acceptable</td><td>Acceptable</td><td>Acceptable</td></tr><tr><td>L≤5.0</td><td>0.05<W≤0.1</td><td>4</td><td>5</td><td>6</td></tr><tr><td>L>5.0</td><td>W>0.1</td><td>0</td><td>0</td><td>0</td></tr></table>	Zone Size (mm)		Acceptable Qty				A+B			C	Length	Width	3.5''~7''	7~10.1''	>10.1''	Ignore	W≤0.05	Acceptable	Acceptable	Acceptable	L≤5.0	0.05<W≤0.1	4	5	6	L>5.0	W>0.1	0	0	0	Minor	
Zone Size (mm)		Acceptable Qty																																
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L>5.0	W>0.1	0	0	0																														
4	Polarizer defect	5.4.1 Polarizer Position (i) Shifting in position should not exceed the glass outline dimension. (ii) Incomplete covering of the viewing area due to shifting is not allowed. 5.4.2 Dirt on polarizer Dirt which can be wiped easily should be acceptable. 5.4.3 Polarizer Dent & Air bubble <table><tr><th colspan="2" rowspan="2">Zone Size(mm)</th><th colspan="3">Acceptable Qty</th></tr><tr><th colspan="3">A+B</th><th rowspan="4">C</th></tr><tr><th colspan="2"></th><th>3.5''~7''</th><th>7~10.1''</th><th>>10.1''</th></tr><tr><td colspan="2">Φ ≤0.2</td><td>Acceptable</td><td>Acceptable</td><td>Acceptable</td></tr><tr><td colspan="2">0.2< Φ ≤0.5</td><td>4</td><td>5</td><td>6</td></tr><tr><td colspan="2">Φ >0.5</td><td>0</td><td>0</td><td>0</td></tr></table> 5.4.4 Polarizer scratch (i) If the polarizer scratch can be seen after cover assembling or in the operating condition, judge by the linear defect of 5.3. (ii)If the polarizer scratch can be seen only in non-operating condition or some special angle, judge by the following: <table><tr><th>Zone</th><th>Acceptable Qty</th></tr></table>	Zone Size(mm)		Acceptable Qty			A+B			C			3.5''~7''	7~10.1''	>10.1''	Φ ≤0.2		Acceptable	Acceptable	Acceptable	0.2< Φ ≤0.5		4	5	6	Φ >0.5		0	0	0	Zone	Acceptable Qty	Minor
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Size (mm)		A+B			C																									
Length	Width	3.5''~7''	7~10.1''	>10.1''																										
Ignore	W≤0.05	Acceptable	Acceptable	Acceptable	Acceptable																									
1.0<L ≤5.0	0.05< W≤0.20	4	5	6																										
L>5.0	W>0.2	0	0	0																										
5	MURA	Using 3% ND filter, it's NG if it can be seen in R,G,B picture.	Minor																											
	White/Black dot (MURA)	Visible under: ND3%; D≤0.15mm, Acceptable; 0.15mm<D≤0.5mm, N≤4; D>0.5mm, Not allowable.																												
6	Glass defect	(i) Crack Cracks are not allowed. 	Minor																											
		(ii) TFT chips on corner  <table><tr><td>X</td><td>Y</td><td>Z</td><td>Acceptable</td></tr><tr><td>≤3.0</td><td>≤3.0</td><td>Not more than the thickness of glass</td><td>N≤3</td></tr></table>	X	Y	Z	Acceptable	≤3.0	≤3.0	Not more than the thickness of glass	N≤3	Minor																			
X	Y	Z	Acceptable																											
≤3.0	≤3.0	Not more than the thickness of glass	N≤3																											

		Chips on the corner of terminal shall not be allowed to extend into the ITO pad or expose perimeter seal.									
		(iii) Usual surface crack  <table border="1"> <thead> <tr> <th>X</th><th>Y</th><th>Z</th><th>Acceptable</th></tr> </thead> <tbody> <tr> <td>≤ 1.5</td><td>≤ 1.5</td><td>Not more than the thickness of glass</td><td>$N \leq 4$</td></tr> </tbody> </table> It is only applicable to the upper glass of LCD.	X	Y	Z	Acceptable	≤ 1.5	≤ 1.5	Not more than the thickness of glass	$N \leq 4$	Minor
X	Y	Z	Acceptable								
≤ 1.5	≤ 1.5	Not more than the thickness of glass	$N \leq 4$								

12.6 Module Cosmetic Criteria

Item No.	Items to be inspected	Inspection Standard	Classification of defects
1	Difference in Spec.	Not allowable	Major
2	Pattern peeling	No substrate pattern peeling and floating	Major
3	Soldering defects	No soldering missing	Major
		No soldering bridge	Major
		No cold soldering	Minor
4	Resist flaw on PCB	Visible copper foil ($\Phi 0.5$ mm or more) on substrate pattern is not allowed	Minor
5	FPC gold finger	No dirt, breaking, oxidation lead to black	Major
6	Backlight plastic frame	No deformation, crack, breaking, backlight positioning column breaking, obvious nick.	Minor
7	Marking printing effect	No dark marking, incomplete, deformation lead to unable to judge	Minor
8	Accretion of metallic Foreign matter	No accretion of metallic foreign matter (Not exceed $\Phi 0.2$ mm)	Minor
9	Stain	No stain to spoil cosmetic badly	Minor

10	Plate discoloring	No plate fading, rusting and discoloring	Minor
11	1. Lead parts	a. Soldering side of PCB Solder to form a 'Filet' all around the lead. Solder should not hide the lead form perfectly.	Minor
		b. Components side(In case of 'Through Hole PCB') Solder to reach the Components side of PCB.	Minor
	2. Flat packages	Either 'Toe'(A) or 'Seal'(B)of the lead to be covered by "Filet". Lead form to be assume over Solder. 	Minor
	3. Chips	$(3/2) H \geq h \geq (1/2) H$ 	Minor
	4. Solder ball/Solder splash	a. The spacing between solder ball and the conductor or solder pad $h \geq 0.13$ mm. The diameter of solder ball $d \leq 0.15$ mm.	Minor
		b. The quantity of solder balls or solder splashes isn't beyond 5 in 600 mm ² .	Minor
		c. Solder balls/Solder splashes do not violate minimum electrical clearance.	Major